

Public Transport For All?

A Multi-Method Spatial Analysis of Transit Accessibility and Socio-Economic Patterns in Stockholm

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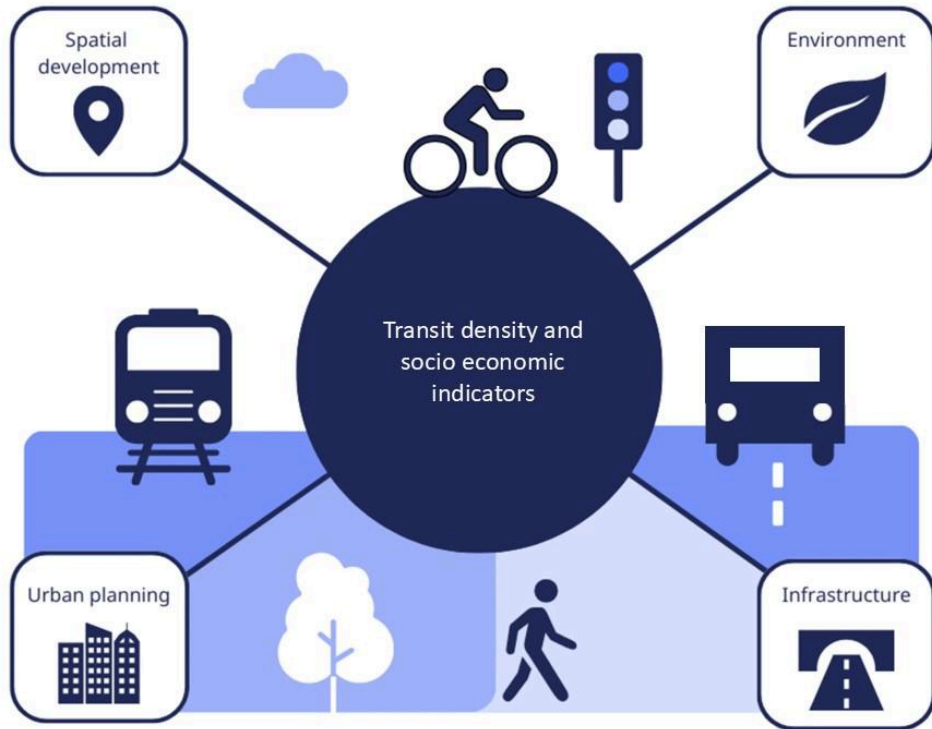


Figure 1: Transit density and socio economic patterns| Source: Adapted from PTV Group. (n.d.). *Transportation planning*. [PTV Group Transportation Planning](#)

Table of contents

1. Introduction	3
1.1 Aim	3
1.2. Research question	3
2. Theoretical Framework	3
2.1 GIS Precedent	4
2.2 The Equity Promise of Public Transport	4
2.2 Income Segregation and the Geography of Stockholm	4
2.3 Who Depends on Transit and What It Actually Costs Them	5
2.4 Evidence of Spatial Inequality in Transit Provision	5
2.5 The Gap This Study Addresses	5
3. Methodology	6
3.1 Data Acquisition	6
3.2 Data Processing	7
3.3 Spatial Autocorrelation and Cluster Analysis (LISA)	15
3.4 Correlation analysis using SPSS	16
3.5 Multivariate Ordinary Least Squares (OLS) regression and spatial diagnostic	17
3.6 Assumptions and delimitations	17
3.7 Final methodological framework	18
4. Results	18
4.1 Spatial Autocorrelation and Cluster Analysis (LISA)	18
4.2 Correlation analysis	25
4.3 Regression analysis	27
5. Discussion	29
6. Future research questions	31
7. Conclusion	31
References:	32
Appendix	33

1. Introduction

Public transport accessibility plays an important role in shaping mobility, connectivity, and access to opportunities within cities. In transit-oriented urban systems such as Stockholm, transport infrastructure is closely linked to urban density, regional connectivity, and patterns of socio-economic development. However, accessibility is rarely distributed evenly across urban areas, making it important to understand how transit patterns relate to broader urban and socio-economic structures.

Stockholm provides an interesting case study due to its extensive public transport network and strongly radial urban structure centred around the inner city. At the same time, the city displays clear socio-spatial differences between central and peripheral districts, raising questions regarding how transit density is distributed across different parts of the municipality.

This report will evaluate how effective spatial cluster analysis and spatial regression analysis are as methods in exploring the spatial distribution of transit density across Stockholm. The study will investigate the limitations and the benefits of these spatial analysis techniques in defining both geographic clustering patterns and the relationships between transit density and selected urban indicators. Ultimately this report will assess how successful these statistical analysis methods are in helping us to better understand how transit-oriented urban structure is shaped across Stockholm.

The analysis is carried out in two stages. First, spatial cluster analysis is used to examine geographic patterns and clustering across Stockholm. Second, statistical and spatial regression analysis are used to further investigate the relationship between transit density and selected urban variables. The study follows a primarily positivist approach, relying on GIS-based spatial analysis and quantitative statistical methods to examine urban accessibility patterns.

The report also explores how spatial and statistical methods can contribute to understanding urban accessibility and transit-oriented urban structure.

1.1 Aim

- To explore how well statistical analysis methods can evaluate how transit is spatially spread in Stockholm.
- To understand whether transit density and socio-economic indicators exhibit spatial clustering and geographic inequality across Stockholm using cluster analysis.
- To examine the extent to which socio-economic indicators and urban centrality influence transit density patterns across Stockholm using regression.

1.2. Research question

How effectively can statistical and spatial analysis methods evaluate the spatial distribution of transit density across DeSO areas in Stockholm, and what insights does each method provide into patterns of accessibility and inequality?

2. Theoretical Framework

This study will apply multiple different types of statistical analysis tools to explore our chosen topic of transport distribution. Cities are not neutral spaces, the opportunities the inhabitants get are influenced by the area they live in and how easily they can move throughout the city. Public transport is something that governs this factor a lot, it sits at the very center of the relationship between the people and their access to the city. Therefore, the distribution of public transport across cities has been studied extensively as it is one of those urban systems that doesn't discriminate on the

basis of income, background, or residential location. Everyone should have equal access to it since it's 'public' transport. But if transit distribution shows signs of favourism towards a group based on residential location and wealth, the more important factor of need takes a back seat. Those who really need it the most and are dependent on it are not considered, this has been the focus of many statistical analysis studies to date.

This study will assess the methods that can be used to investigate whether this tension really exists in Stockholm. It asks whether lower-income areas systematically have worse public transportation than higher-income areas and how the chosen statistical methods can be used to reveal the answers to these questions in this context.

2.1 GIS Precedent

GIS can and has been used in many studies to allow for the exploration of the interaction between the topics of public transport and spatial distribution. In a "European Transport Research Review" paper from 2012, GIS together with global and local spatial autocorrelation measures and regression models, were used to determine whether trip patterns exhibit spatial dependence and clustering (Mazzulla & Forciniti, 2012). The use of GIS in this case revealed significant concentrations of high and low trip areas, demonstrating that neighbouring zones tend to have similar trip-generation characteristics rather than being randomly distributed (Mazzulla & Forciniti, 2012).

ESDA techniques have increasingly been applied to assess the spatial distribution and accessibility of public transport services. For example, when public transport accessibility was mapped in Jakarta in India, LISA and ESDA cluster analysis was applied as part of this study to identify clusters of high and low public transport accessibility across the city (Karly et al., 2026).

Similar approaches have been used in urban transport studies to investigate trip distributions, transit accessibility, and service coverage, demonstrating the value of ESDA for uncovering spatial dependence and informing transport planning decisions. The precedent set by these studies informs our own study to evaluate the effectiveness of various statistical models in analysing the spatial correlation between public transport distribution across the city of Stockholm.

2.2 The Equity Promise of Public Transport

Access to public transport has always been a social issue. Subsidized transit rates are based not only on environmental or efficiency reasons but also on supporting people who don't have access to private transportation, giving them access to the city. (Lucas, 2012)

Researchers have recognized that when this fails, the question is not just about inconvenience then, but inadequate access to transport creates a poverty trap. Bad connectivity doesn't just reflect disadvantage, but it deepens it. When residents cannot reliably reach work, healthcare, or education, their ability to improve their circumstances is structurally constrained. Public transport, in this sense, is not a luxury. It is infrastructure for social mobility.

This matters for Stockholm because the city presents itself as a model of efficient and inclusive transit. The regional transport authority frames public transit as a tool for sustainable urban development and equal access. Yet the distribution of that access across income groups is precisely what this study puts under scrutiny.

2.2 Income Segregation and the Geography of Stockholm

Understanding whether transit serves all income groups equally requires first understanding where different income groups live. In Stockholm, this has been actively shaped by decades of housing policy and market forces. Large-scale conversions of rental housing to cooperative ownership have concentrated wealthier households centrally while pushing low income renters toward the suburbs. A limited social housing sector combined with a hard to access housing market has driven low income groups into suburban multifamily housing, particularly in the southwest and northwest where poverty concentration is highest.

This process of residential sorting is not incidental. It is the structural outcome of policy decisions that prioritised the protection and enhancement of the inner city while relocating social challenges - including poverty, unemployment and housing vulnerability - to the suburban periphery. As our own experience arriving in Stockholm confirmed, the further one travels from the city centre the more visible these concentrations of disadvantage become. And as this study will show, the spatial distribution of income across Stockholm's DeSO areas is neither random nor confined to the south of the city as one might initially assume. The thirty lowest income areas are spread across the city with particular concentration in the northwest and southwest - a finding that itself challenges common assumptions about Stockholm's urban geography.

2.3 Who Depends on Transit and What It Actually Costs Them

The question of who uses public transport and who depends on it are not the same. Across income groups, transit use can look similar in aggregate while masking a fundamental difference in dependency. Wealthier residents choose public transport. Lower income residents frequently have no choice.

Analysis of Stockholm smart card data during the pandemic showed that higher income residents stopped using public transport at far higher rates while lower income residents kept travelling at much higher rates. This data makes the dependency relationship explicit. When public transport became a perceived health risk, those who could avoid it did. Those who could not - primarily lower income residents without access to private vehicles or remote working - kept using it regardless. Transit is not a preference for these communities. It is the only viable means of accessing the city. (Almlöf et al. 2021)

Yet the system does not reflect this dependency in how it prices or distributes its services. Low income individuals in Sweden cannot afford monthly travelcards and buy single tickets instead, meaning they pay more per trip than affluent individuals. In Stockholm this overpayment by lower income riders has been estimated at SEK 250 million per year - a substantial sum that represents a systematic failure of the equity argument used to justify transit subsidies. The people most reliant on the system are contributing disproportionately to its revenue while receiving, as this study will examine, potentially less in return.

2.4 Evidence of Spatial Inequality in Transit Provision

The financial dimension of inequality is compounded by its spatial dimension. Research on Swedish cities suggests that the areas of greatest need are also, in many cases, the areas of weakest provision. Studies using GTFS data and DeSO areas find that high social deprivation and low transit provision tend to coincide spatially, with this pattern visible in Stockholm's suburban fringe.

Further evidence comes from the structure of transit subsidies themselves. The subsidy per trip in peripheral Stockholm is ten times higher than in the regional core - not because those areas receive generous service but because services are so infrequent that each trip costs more to provide. This is a critical distinction that aggregate funding figures obscure. High subsidy per trip in outer areas does not mean investment. It means inefficiency. Sparse, infrequent services cost more per journey precisely because they carry fewer passengers. The subsidy structure reveals the problem rather than correcting it. (Bondemark et al. ,2021)

Research on urban inequality in Stockholm finds that social segregation is the primary driver of inequality in access to urban services. Transit is one of the most consequential of those services. Where residential segregation is deepest, access to the mobility infrastructure that enables opportunity is weakest.

2.5 The Gap This Study Addresses

The literature reviewed here establishes that transit inequality in Stockholm is real, costly and spatially structured. Lower-income residents are concentrated in specific outer areas, depend on transit more than anyone else and pay more to access a system that research suggests is weaker in the areas they inhabit. This is not a new observation - but it has not been rigorously tested at the fine grained DeSO level using a combination of spatial and statistical methods

that can quantify the relationship, identify where it clusters and evaluate which analytical approaches are most effective at revealing it.

That is the gap this study fills. By applying OLS regression and LISA cluster analysis to DeSO level data across Stockholm municipality, this report contributes both a substantive finding on transit distribution and a methodological evaluation of the tools used to measure it.

3. Methodology

3.1 Data Acquisition

Data for this study were collected at the DeSO level for Stockholm municipality in order to compare public transport accessibility between lower- and higher-income areas (Statistics Sweden, n.d.). DeSO here refers to Demographic Statistics Areas (Demografiska statistikområden), which are small-area geographic units defined by Statistics Sweden to present detailed socio-economic and demographic statistics at a fine spatial scale; these units are designed to be relatively homogeneous in terms of population and socio-economic characteristics and are used throughout Stockholm and other Swedish municipalities, with this definition of DeSO followed in all subsequent mentions in the text.

The main spatial unit of analysis is the DeSO polygon, which allows the study to link neighbourhood-scale socioeconomic information with transport accessibility data (Boverket, 2021). The selection was limited to contain only DeSO that are spatially located within Stockholm municipality by filtering, leaving only DeSO codes that begin with 0180 (code for Stockholm). This defined the study area, shown in Figure 2.

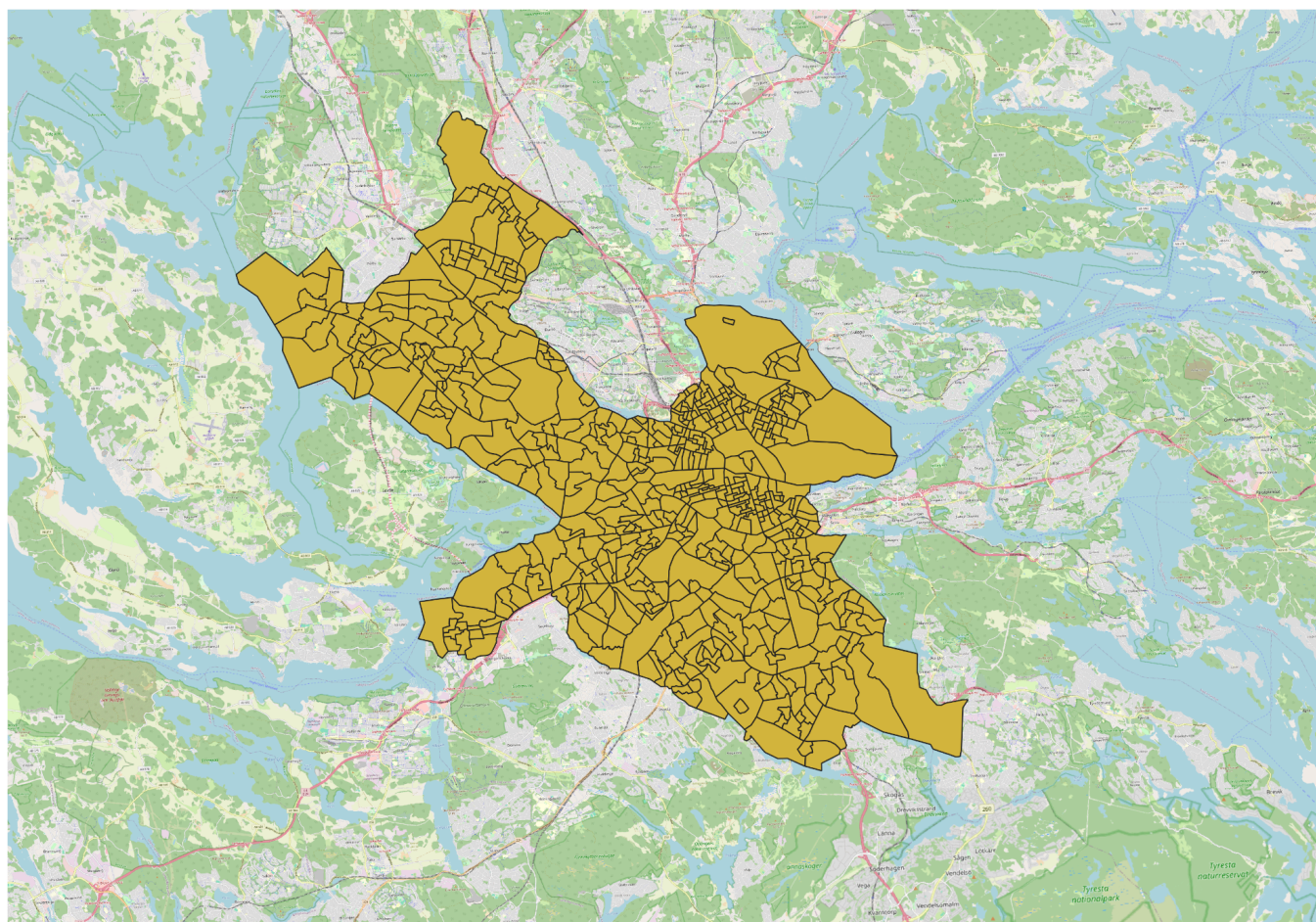


Figure 2: Study area - Stockholm municipality | Source: Authors

Household income data were obtained from Statistics Sweden (SCB), as this is the main source for small-area income data in Sweden (Statistics Sweden, n.d.). Additional variables were collected for each DeSO area, including car

ownership, age structure, and population density, to examine whether these factors are related to differences in public transport accessibility (Boverket, 2021; Stockholm Open Data, n.d.).

Public transport data were collected using stop location data and transit departure data for Stockholm (Trafiklab, n.d.). Stop locations were used to measure how many public transport stops were available within a 400-metre radius of each DeSO area, while transit departures were used to measure the level of service provision (Trafiklab, n.d.). DeSO boundary polygons were also collected to serve as the spatial basis for the analysis (Statistics Sweden, n.d.; Stockholm Open Data, n.d.).

Variable	Description	Source	Purpose
Household income	Average annual income per DeSO area	Statistics Sweden (SCB, n.d.)(Statistics Sweden, n.d.-a)	Deriving the independent variable- Income per person per DeSO
Car ownership	Share or number of households with at least one car per DeSO area	Statistics Sweden (SCB, n.d.)	Deriving the independent variable
Population density	Population per km ² in each DeSO area	Statistics Sweden (SCB, n.d.)	Deriving the dependent variable- Transit density(<i>Transit Stops within 400m walking distance per person per DeSO</i>)
Transit stops (400 m)	Number of public transport stops within 400 m of each DeSO area	GFTS (Stockholm transport data)(Trafiklab, n.d.)	Deriving the dependent variable- Transit density(<i>Transit Stops within 400m walking distance per person per DeSO</i>)
Transit departures	Total number of departures per DeSO area per time period	GFTS (Stockholm transport data)(Trafiklab, n.d.)	Data exploration
DeSO boundary polygons	Administrative boundaries for Demographic Statistical Areas in Stockholm	SCB / Stockholm Open Data(Statistics Sweden, n.d.-b)	Base data

3.2 Data Processing

After the data were acquired and harmonised, they were processed into a format suitable for spatial and statistical analysis. This included projecting all datasets into a common coordinate reference system, checking for missing values and outliers, and merging the socioeconomic variables (income, car ownership and population density) with the public transport accessibility measures at the DeSO level.

Initial data processing and spatial preparation were carried out in QGIS. In addition to calculating public transport accessibility, socioeconomic datasets such as income and car ownership were imported, cleaned, and joined to the

DeSO spatial boundaries within QGIS. Attribute tables were checked for inconsistencies, and the variables were spatially linked to the corresponding DeSO polygons to create a unified spatial dataset. QGIS was also used to calculate the number of public transport stops within a 400 m radius of each DeSO area and to prepare the final dataset for statistical analysis.

Subsequent spatial statistical analysis was conducted in GeoDa. GeoDa was used to perform the exploratory spatial data analysis, including Local Indicators of Spatial Association (LISA) cluster analysis, and to estimate Ordinary Least Squares (OLS) regression models examining the relationship between public transport accessibility and the selected socioeconomic variables.

3.2.1 Selection of variables

These variables are chosen because they map the relationship between a neighborhood's physical access to transit and its socio-economic vulnerability, revealing whether Stockholm's transport network structurally supports or isolates marginalized communities.

3.2.2 Independent variables

To comprehensively evaluate the landscape of transit equity and spatial accessibility across Stockholm's DeSO zones, this study establishes a specialized vector of independent variables. For the scope of this primary analysis, three core socio-economic and spatial indicators have been selected:

Income

Income measures socio-economic vulnerability to operationalize vertical equity, as lower-income groups are inherently more transit-dependent due to the financial barriers of private transport.(Almlöf et al., 2021) In Stockholm, tracking this at the neighborhood level helps evaluate whether transit supply privileges affluent areas or supports marginalized suburban districts.

Distance from central business district

This variable serves as a vital geometric control to account for the highly monocentric, radial nature of Stockholm's transport network. Including it allows the model to separate normal center-periphery infrastructure degradation from active socio-spatial inequities.

Car ownership

Car ownership rates function as a direct behavioral proxy to differentiate between lifestyle choices and forced public transport dependency(Lucas, 2012). It isolates *transit deserts* by identifying low-income peripheral zones where residents lack both a private vehicle and adequate public transport access.

While it is recognized that transport equity is multifaceted and that multiple other indicators, such as education levels, employment rates, demographics, and country of origin remain valuable dimensions to be explored in later stages of research. This chosen configuration offers a highly focused and robust framework for isolating immediate socio-spatial disparities.

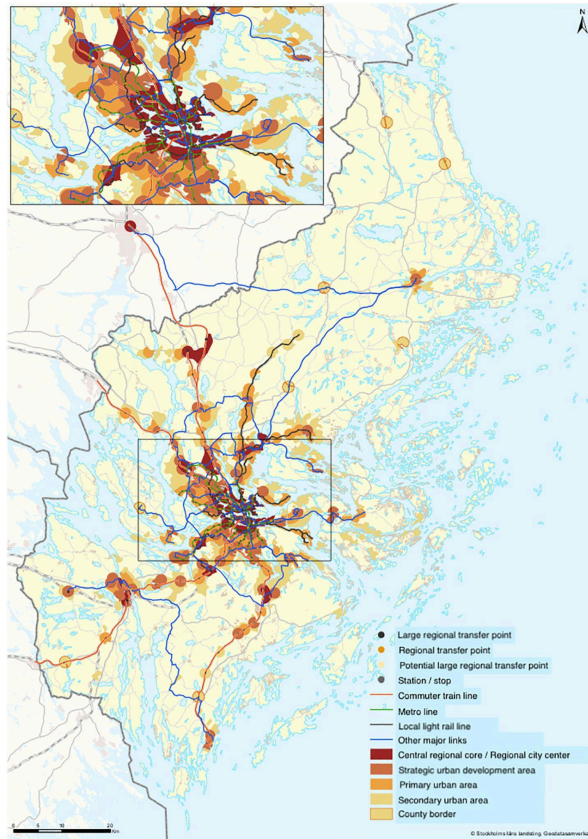


Figure 3: Geographical overview of Stockholm area public transport network(Radial pattern)| Source: (Region Stockholm [RS], 2018)

Dependent variable	Description
Transit density	Number of multi-modal transit stops (bus, Tunnelbana, tram/light rail) within a 400m walking buffer per 1,000 residents per DeSO.
Independent variable (Socio economic indicators)	Description
Income	Average income per person per DeSO.
Distance from central business district(T centralen)	Geographic Euclidean or network distance measured from the centroid of each DeSO to T-Centralen.
Car ownership	Vehicle registration rate calculated as total passenger cars per capita per DeSO.
Age structure	Dominate age per DeSO

3.2.3 Dependent Variable Specification: Transit density

This variable calculates spatial accessibility within an established five-minute walking buffer (400m) to evaluate localized transit infrastructure provision. Incorporating population into this metric is crucial because a simple spatial count of transit stops fails to capture the actual pressure on infrastructure. Standardizing this density per 1,000 residents accounts for local residential density, shifting the focus from mere geographic presence to true

supply-versus-demand dynamics. The adjustment prevents the model from overestimating access in sparsely populated areas. It also ensures that high-density residential zones, where ridership demand and transit reliance are highest are accurately evaluated for resource adequacy .

Number of stops in 400m walking radius per DeSo centroid

The first accessibility indicator measures the availability of public transport stops within a reasonable walking distance. A threshold of 400 metres was adopted, as it is commonly used in transport planning studies to represent an acceptable walking distance to public transport.

DeSO polygons were converted into centroid point features in QGIS. Each centroid was assumed to represent the geographic centre of its corresponding statistical area. A 400-metre Euclidean buffer was then generated around each centroid to represent the area that could potentially be reached within walking distance from the centre of the DeSO.

Transit stop locations were obtained from GTFS datasets (Trafiklab, n.d.) and imported into QGIS as point features. Using the Count Points in Polygon tool, the number of transit stops located within each centroid buffer was calculated. This produced a measure of the number of public transport stops accessible within a 400-metre radius of the representative location of each DeSO.

The use of centroids provides a simplified representation of accessibility at the statistical-area level. While this approach does not capture variations in population distribution within a DeSO or actual walking routes along the street network, it offers a consistent and computationally efficient method for comparing accessibility across the study area.

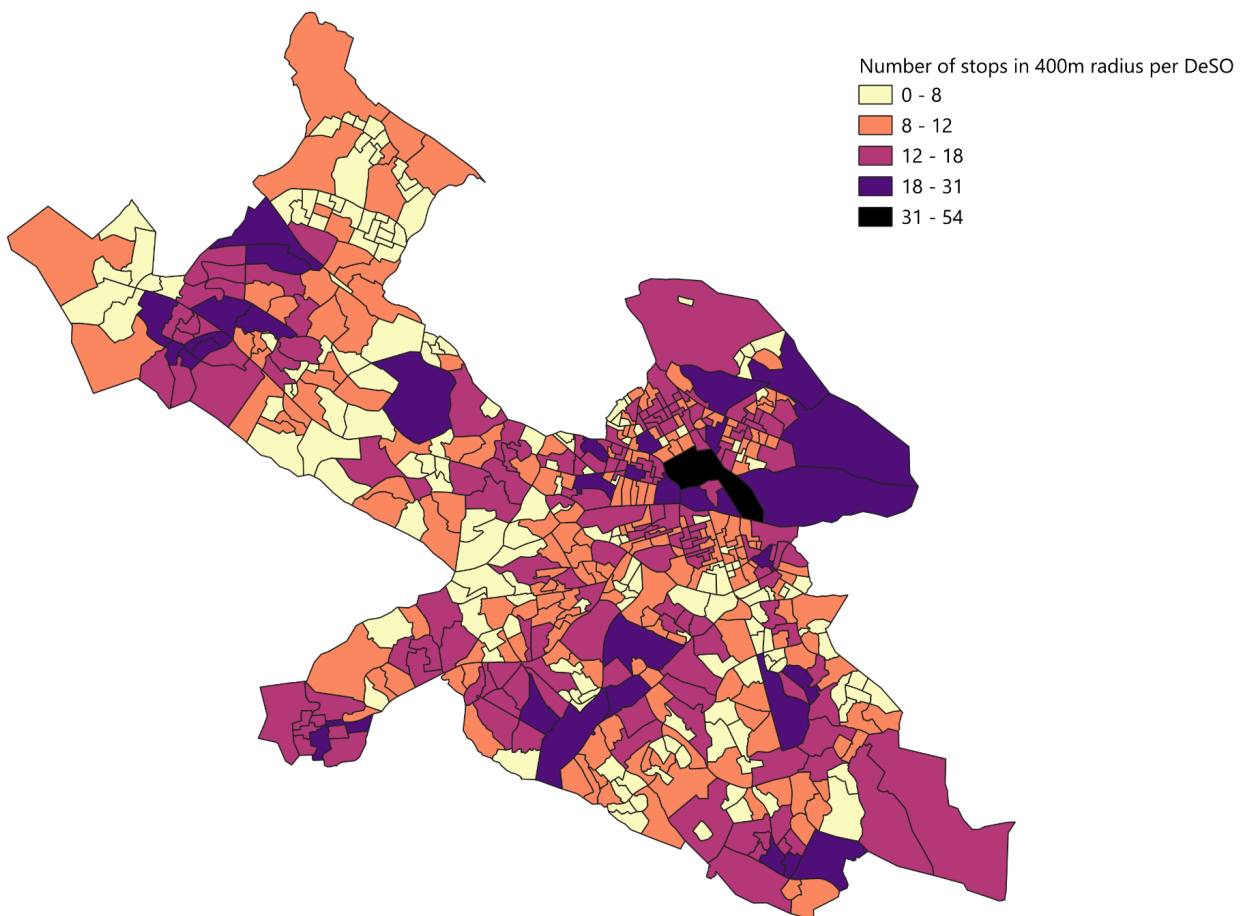


Figure 4: Map showing number of stops(all modes) within 400m radius per DeSO centroid| Source: Authors

Number of stops in 400m walking radius per 1000 people per DeSo

Analyzing absolute stop counts introduces severe scale bias, as a sprawling or low-density zone might mathematically register a high number of stops without reflecting genuine service density or spatial equity for its residents. To control for these variations in district size and demographic density, the raw data was normalized by dividing the total number of stops by the local population. Multiplying this quotient by 1,000 provides a standardized metric: the number of public transport stops available per 1,000 residents. This normalization provides population-adjusted transit density for each DeSO (Measured from its centroid).

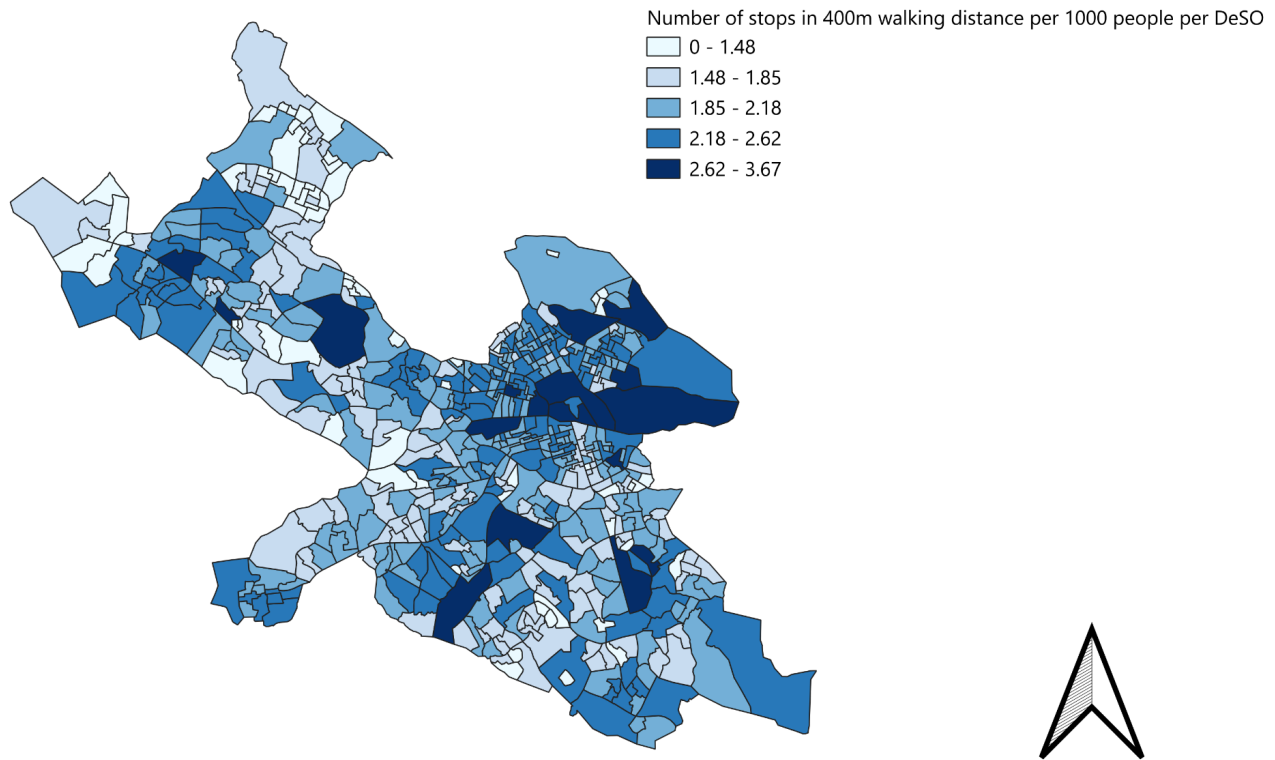


Figure 5: Map showing number of stops(all modes) within 400m radius per 1000 residents per DeSO(Logarithmic function)| Source: Authors

While per-capita normalization addresses demographic scale biases, the resulting distribution retains a severe right skew (See figure 6). This asymmetry is driven by extreme statistical outliers, specifically, specialized commercial zones, industrial parks, or central transit corridors that maintain low residential populations but a high concentration of transit infrastructure. In a standard OLS framework, these high-value outliers disproportionately distort the calculation of the regression line due to the squaring of residual distances. Furthermore, this distribution exhibits severe heteroscedasticity, violating the core regression assumption of constant variance.

Distributional correction via logarithmic transformation of transit density

To resolve these right-skewed distributional flaws in the raw metrics, a natural logarithmic transformation ($\ln(x + 1)$) was applied to the per-capita transit density metric. The $+1$ constant adjustment is methodologically vital, as it mathematically accommodates extreme *transit deserts*, i.e. DeSO zones with a raw stop count of zero, by binding them safely to a transformed value of exactly zero ($\ln(0+1)=0$).

As demonstrated in the transformed histogram, the logarithmic function(See figure 6) compresses the extreme right tail while expanding the variance among lower-service cohorts. While this expansion results in a prominent negative (left) skewness where severe transit deserts anchor the absolute baseline of the x-axis, the overall transformation significantly stabilizes the variance and approaches a structure suitable for robust Ordinary Least Squares (OLS) regression modeling.

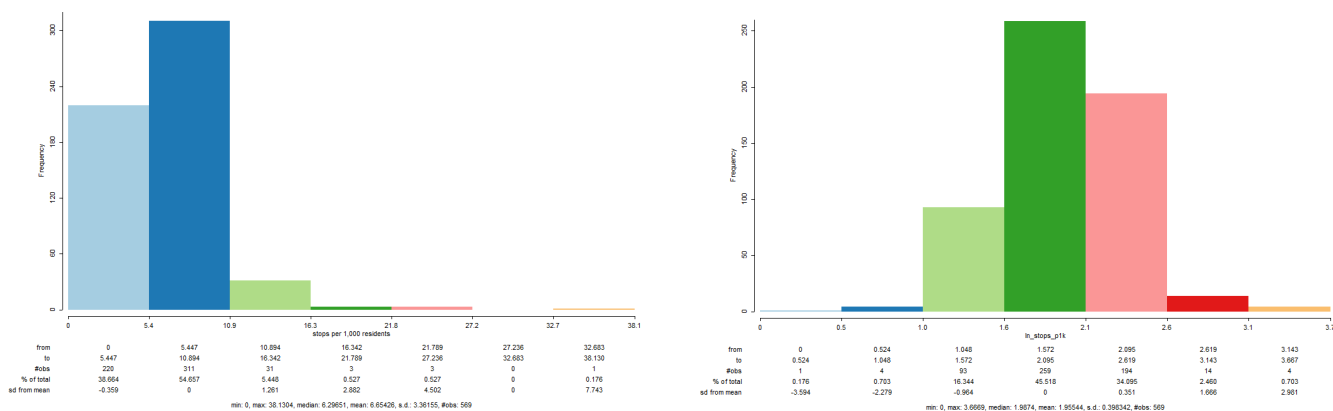


Figure 6: Comparison of the raw count of stops in 400m distance per 1000 residents and the logarithmic function of the number of stops in 400m walking radius per 1000 people per DeSO| Authors

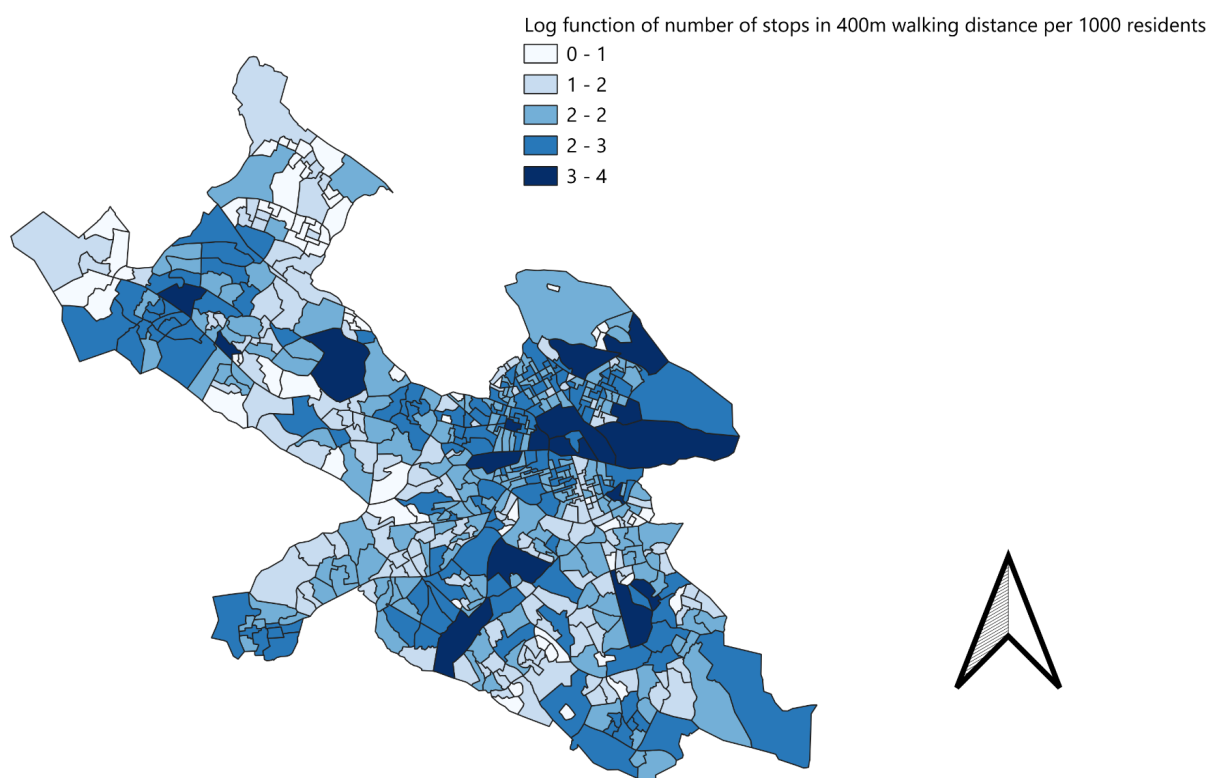


Figure 7: Log function of number of stops in 400m walking distance per 1000 residents| Source: Authors

The log-transformed value ($\ln(\text{Transit Density} + 1)$) is the final version of the dependent variable.

3.2.4 Definition of independent variable

Income per person per DeSO

The raw socioeconomic dataset, containing mean income per person, was obtained as a tabular format linked to unique DeSO identification codes. The foundational GIS layer consisting of Stockholm County's 569 DeSO polygon boundaries was imported into QGIS. Using the native Join Attributes by Field Value tool, the tabular income data was structurally coupled with the spatial layer using the unique DeSO ID code as the common key attribute. This successfully assigned localized attribute profiles to the explicit physical geometries of each statistical area.

Once the table join was successfully compiled, exploratory data analysis was conducted within *GeoDa*. An initial histogram of the raw income per person variable was generated to observe its statistical distribution across Stockholm (See figure 8). The histogram revealed a highly right-skewed (positively skewed) profile, demonstrating that a small percentage of highly affluent neighborhoods skewed the regional data distribution.

Distributional correction via logarithmic transformation

Because highly skewed data violates the parametric distribution assumptions required for localized spatial statistics, a natural logarithmic transformation ($\ln(x)$) was applied directly to the income attribute field using the QGIS Field Calculator. This compressed the long right-hand tail of the distribution, resulting in a symmetrical, normal bell-curve structure. Transforming the variable within QGIS in this manner was an essential methodological prerequisite before performing the Local Indicators of Spatial Association (LISA) analysis, as it prevents extreme outlier values from distorting spatial autocorrelation calculations and local cluster significance (*p-values*).

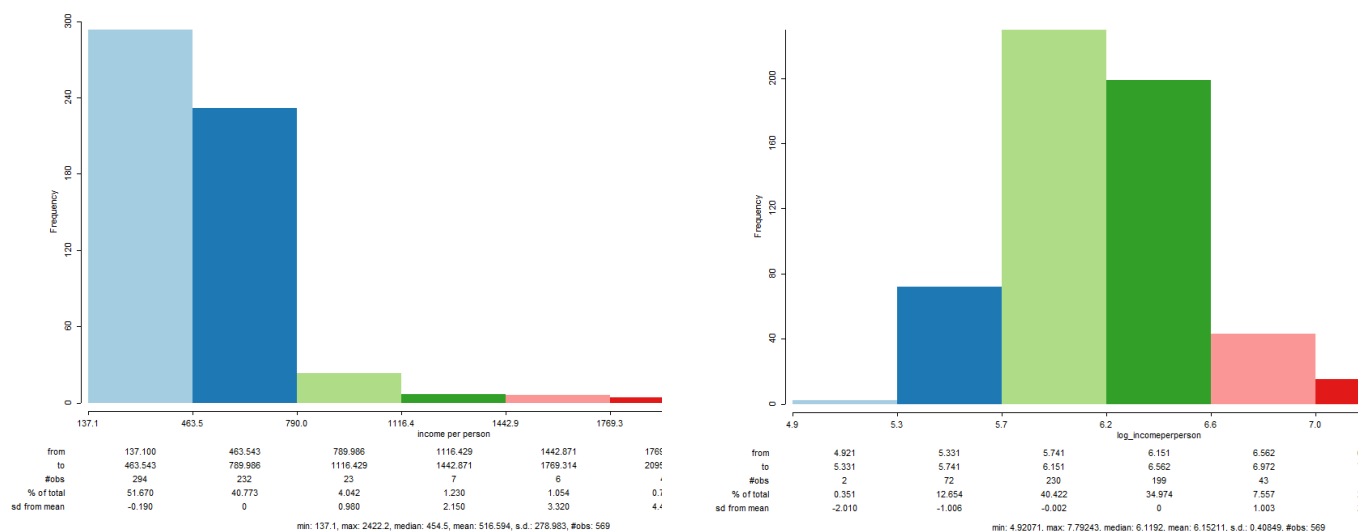


Figure 8: Comparison of the histograms showing distribution of income person per DeSO and the logarithmic function of income person per DeSO| Authors

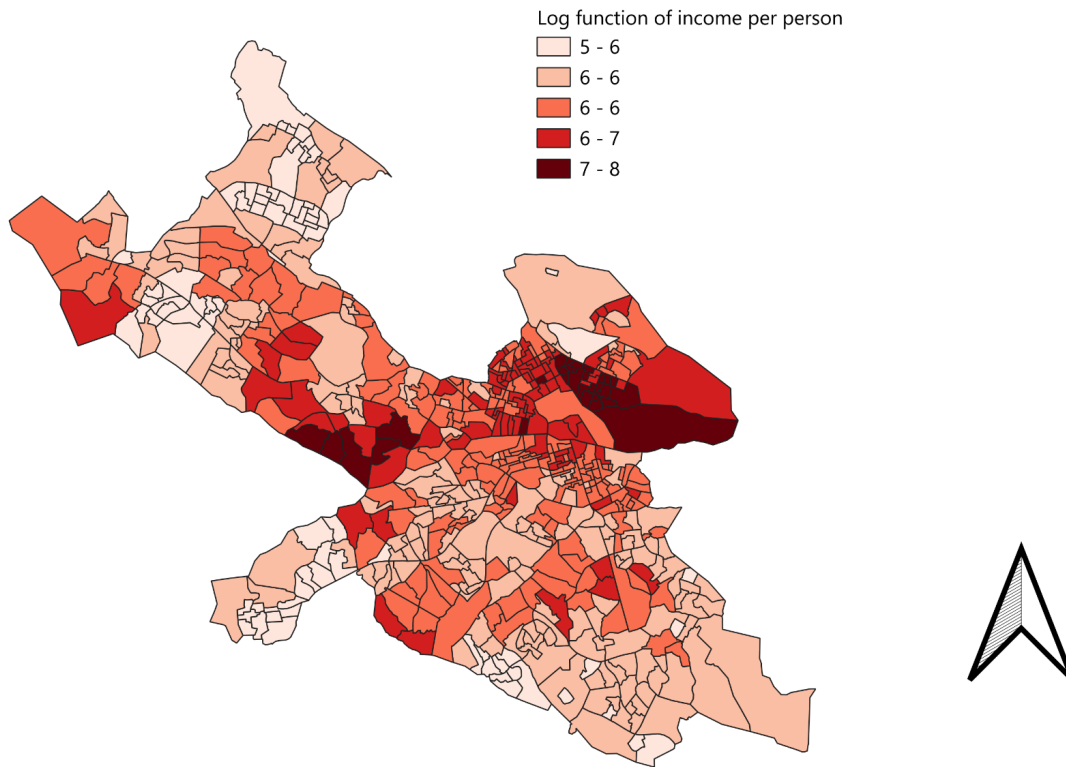


Figure 9 : Log function of income per person per DeSO| Source: Authors

Distance from central business district and car ownership per person

The distance from CBD variable (`dist_cbd`) was generated in QGIS to measure the relative proximity of each DeSO area to Stockholm's urban core. T-Centralen was selected as the CBD reference point due to its role as Stockholm's primary transport and commercial hub.

The centroid of each DeSO polygon was first generated to represent the approximate centre point of each statistical area. Euclidean distance was then calculated between the centroid of every DeSO and the T-Centralen reference point using QGIS distance analysis tools. The output produced a continuous distance variable measured in kilometres for all 569 DeSO units.

The variable was included within the regression analysis to account for the effect of urban centrality on transit density. Since Stockholm's transport infrastructure is highly concentrated around the inner city and major radial transit corridors, distance from the CBD was expected to influence transit stop density across the municipality. Including the variable also allowed us to examine whether transit density patterns were shaped more strongly by socio-economic indicators such as income and car ownership, or by broader geographic structure and proximity to the urban core. Another variable, car ownership per person (`cars_per_person`), was generated using data obtained from Statistics Sweden (SCB). The dataset was joined to the DeSO boundary layer in QGIS using the DeSO code, and a new field was calculated by dividing the total number of registered passenger cars by the population of each DeSO area. The resulting standardized variable was later used in the regression analysis.

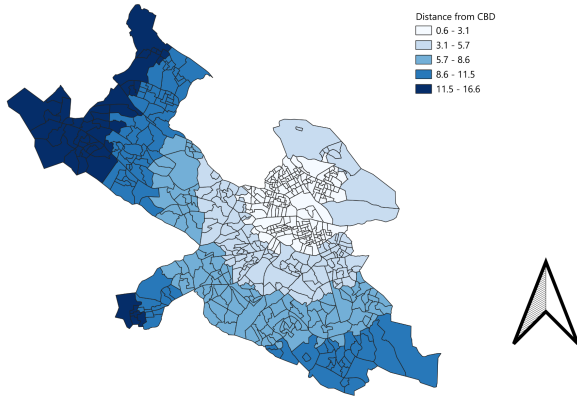


Figure 10: Distance from CBD| Source: Author

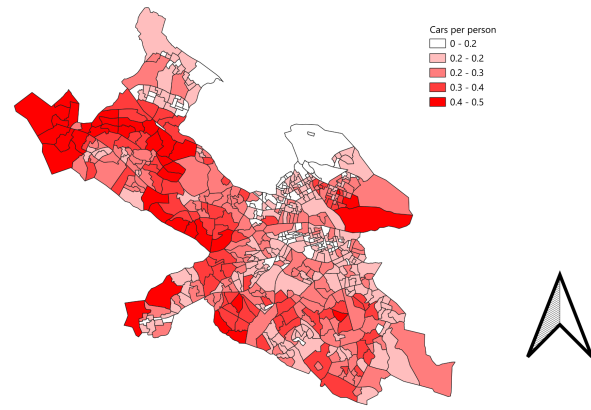


Figure 11: Car ownership per person| Source: Author

Dominant age Group per DeSO

To capture the demographic profile of each Demographic Statistical Area (DeSO) in Stockholm, a dominant age group variable (dom_age_num) was utilized. Raw population data categorized by age brackets was analyzed to identify the cohort with the highest population share within each individual DeSO. These dominant age brackets were then converted into numeric values to enable their inclusion as a continuous predictor within the regression models.

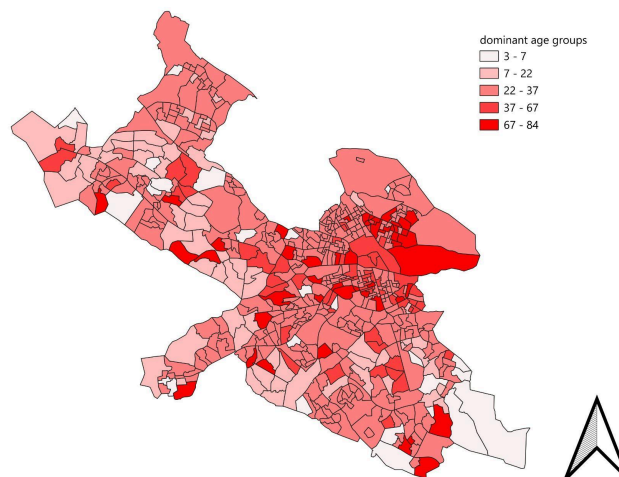


Figure 12: Dominant age group perv DeSO| Source: Author

3.3 Spatial Autocorrelation and Cluster Analysis (LISA)

The next stage of the methodology involves spatial statistical testing to move beyond visual observation and establish the mathematical significance of transit patterns. The purpose of this step is to identify spatial clusters of transit density and examine whether high or low density areas cluster geographically across Stockholm.

To do this, the study directly compares the transformed transit density variable with the transformed mean income variable. Comparing these two metrics is a conceptual choice and it allows the study to test for spatial mismatch and determine if the transport network follows principles of social equity. By analyzing them together, the model can mathematically prove whether low-income populations are trapped in *transit deserts* (areas with both low income and low transit access) or if wealthy neighborhoods receive an unfair share of transport infrastructure.

This analysis was conducted using *GeoDa (version 1.22)*, an open-source software specialized in Exploratory Spatial Data Analysis (ESDA) and spatial econometrics.

*The log-transformed value of the number of transit stops within a 400-metre walking distance per 1,000 residents for each DeSO is referred to as **ln_stops_p1k** throughout this study. The log-transformed value of the mean income per person for each DeSO is similarly referred to as **log_incomeperperson**.*

3.3.1 Definition of Spatial Weights

To quantify the *neighborhood relationships* between the 500+ DeSO areas in Stockholm, a spatial weights matrix was constructed. A Queen's Contiguity criterion (1 st order) was selected. Under this definition, two DeSO areas are considered neighbors if they share a common boundary or a single vertex (corner). This approach is suitable for urban administrative units like DeSO, as it accounts for the continuous nature of the urban fabric and the potential for transit spillover effects across neighboring polygons.

3.3.2 Global Moran's I for transit density and income

The initial step in the spatial analysis evaluates Stockholm County as a whole using Global Moran's I to detect regional patterns of spatial dependence. The Global Moran's I statistic measures overall spatial autocorrelation, outputting a value ranging from -1 (perfect dispersion) to +1 (perfect clustering), where a value near 0 indicates a spatially random distribution. This study utilizes both univariate and bivariate applications.

Univariate Global Moran's I: This test was executed to evaluate the geographic distribution of a single variable against itself across space. It was applied independently to *ln_stops_p1k* and *log_incomeperperson* to verify whether transit access points and neighborhood income levels are structurally clustered across the region or merely randomly distributed.

3.3.3 Local Indicators of Spatial Association (LISA)

Because global statistics summarize the entire study area into a single value, they can mask localized anomalies, pockets of inequality, or shifting spatial regimes. To capture these localized patterns, Local Indicators of Spatial Association (LISA) framework was implemented in both its univariate and bivariate forms:

Univariate LISA:

This model measures the local spatial autocorrelation of a single variable, identifying statistically significant local spatial clusters (hotspots and coldspots) of transit density (*ln_stops_p1k*) on its own, independent of socioeconomic data.

Bivariate LISA (BiLISA):

This model cross-references two different variables to analyze local spatial correlation. It evaluates the spatial relationship between the local independent variable (*log_incomeperperson*) at a specific DeSO location and the average value of the dependent variable (*ln_stops_p1k*) in the surrounding neighborhood features (the spatial lag). This technique provides an empirical test for spatial mismatch and transport equity by locating specific neighborhoods where income disparities match infrastructure discrepancies.

3.4 Correlation analysis using SPSS

Prior to regression modelling, Spearman rank correlation analysis was used as an exploratory technique to examine the relationship between transit stop density and the explanatory variables, including age structure, income, car ownership, and distance to the CBD. Spearman's coefficient was selected because it is a non-parametric measure that does not

require normally distributed data and is suitable for identifying monotonic relationships between socio-spatial variables. The results provided an initial indication of the relative strength of associations and informed the selection and interpretation of variables in the subsequent regression analyses. In addition, correlations among the explanatory variables were examined to identify potential multicollinearity concerns before model estimation.

3.5 Multivariate Ordinary Least Squares (OLS) regression and spatial diagnostic

The results of Spearman correlation analysis were used to inform variable selection for the OLS model by identifying variables that exhibited weak or no association with the dependent variable and by flagging pairs of independent variables with high correlation, which may indicate redundancy and inflate variance in parameter estimates. Following this diagnostic step, OLS regression was used to estimate the linear relationship between the dependent variable (*ln_stops_plk*) and the selected explanatory variables, under the standard assumptions of linearity, independence, homoscedasticity, and absence of perfect multicollinearity.

3.5.1 Spatial Diagnostic Test Framework & Decision Rules

A baseline Classic OLS regression was executed to establish an empirical benchmark and diagnose spatial misspecification. To determine the correct spatial model specification, the Lagrange Multiplier (LM) diagnostics were evaluated based on the following decision rule:

Spatial Lag Model (SLM): Selected if the Robust LM (Lag) test is significant, assuming that the dependent variable in one spatial unit directly influences neighboring units through direct endogenous spillover effects.

Spatial Error Model (SEM): Selected if the Robust LM (Error) test is significant, assuming that spatial dependence is driven by omitted variables with an inherent geographic structure rather than direct endogenous spillovers.

3.6 Assumptions and delimitations

- In this study transit accessibility was measured primarily through transit stop density rather than actual service accessibility. While stop density provides a useful spatial approximation of transit provision, it does not account for important factors such as service frequency, waiting times, reliability, route connectivity, or travel time.
- Transit density was simplified to absolute point counts of stops per DeSO. All modes were treated equally, ignoring distinct operational differences, capacities, and catchment areas of metro lines versus bus networks.
- The study assumes that aggregating infrastructure data into static DeSO polygons accurately captures real-world passenger access, and that a first-order Queen contiguity matrix sufficiently represents neighbor interactions.
- The model relies on straight-line Euclidean distance to the CBD to represent radial network decay, ignoring actual street networks, topography, or true pedestrian routing.
- Scope was strictly delimited to Stockholm municipality's administrative borders, omitting cross-border interactions and connectivity from adjacent DeSOs.
- Complex built-environment attributes including physical topography, commercial/residential zoning layers, and housing market indicators like average rental prices and property values were excluded due to project timeframe constraints.

3.7 Final methodological framework



Figure 13: Workflow of analysis| Source: Author

4. Results

4.1 Spatial Autocorrelation and Cluster Analysis (LISA)

Before executing global and local spatial statistics, a connectivity diagnostic of the generated first-order Queen contiguity spatial weights matrix was conducted to ensure the structural integrity of the neighbor definitions.

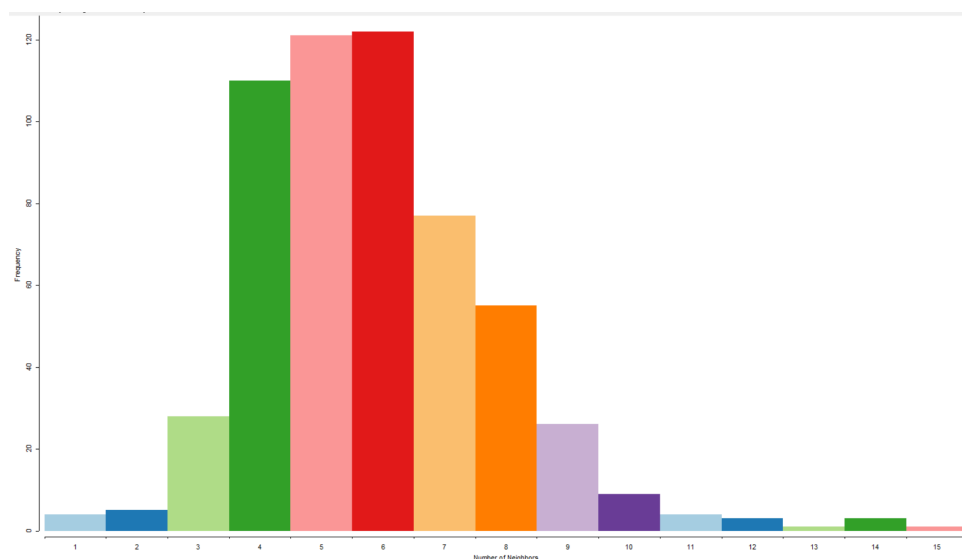


Figure 14: Connectivity histogram of the spatial weights matrix. Source: Author.

The diagnostic histogram (Figure 13) confirms a robust and balanced spatial structure across Stockholm County's 500+ DeSO units. The connectivity profile follows a near-normal bell curve distribution, characterized by a mean connectivity of approximately 6 neighbors per DeSO unit. The structural consistency of this matrix verifies that the neighborhood definitions are clear and unbiased, preventing artificial spatial anomalies from skewing subsequent autocorrelation modeling.

4.1.1 Univariate Moran's I for transit density (*ln_stops_p1k*)

To evaluate whether public transport density is systemically structured across the metropolitan area or merely distributed by random chance, a univariate Global Moran's I test was executed on the log-transformed transit density variable (*ln_stops_p1k*).

The spatial analysis yielded a moderate-to-strong positive regional spatial autocorrelation, returning an observed **Global Moran's I value of 0.3940**. To assess the statistical validity of this metric and account for spatial baseline assumptions, a conditional permutation procedure utilizing 999 randomizations was executed to construct an empirical reference distribution under the null hypothesis of Complete Spatial Randomness (CSR). [Insert

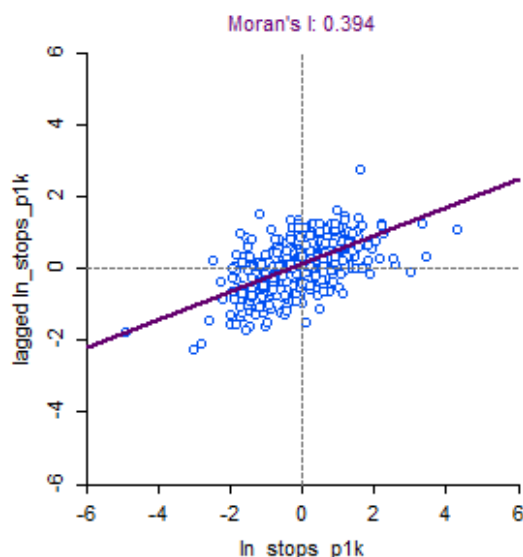
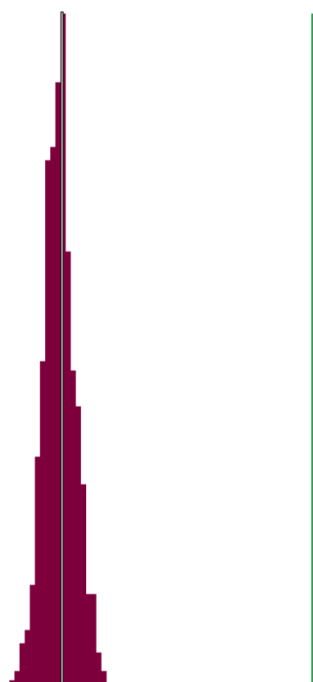


Figure 15: Global Moran's I scatter plot for public transit density (*ln_stops_p1k*). Source: Author.

permutations: 999
pseudo p-value: 0.001000



I: 0.3940 E[]: -0.0018 mean: -0.0017 sd: 0.0241 z-value: 16.4223

Figure 16: Permutation reference distribution (999 randomizations) for univariate Global Moran's I of public transit density (*ln_stops_p1k*). Source: Author.

The empirical results from this permutation test (Figure x) generated a standard score (**z-score**) of **16.0082** and a highly significant pseudo **p-value of 0.001**. As illustrated by the reference distribution histogram, the 999 randomized iterations form a tight, normal bell curve centered around the expected baseline value of spatial randomness:

4.1.2 Univariate Global Moran's I for income per person per DeSO(log_incomeperperson)

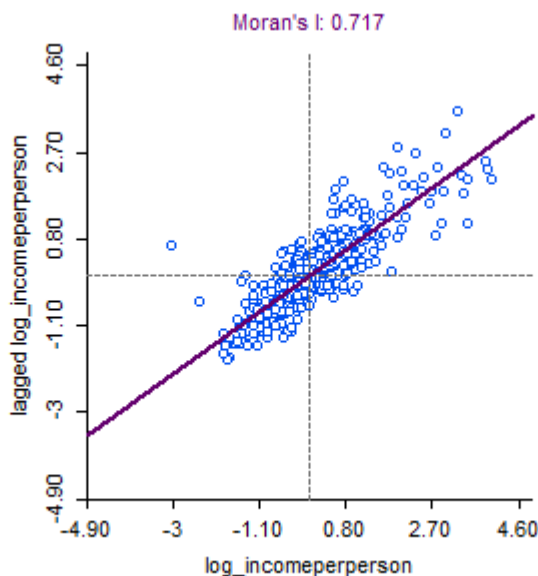


Figure 17: Global Moran's I scatter plot for income per person per DeSO (log_incomeperperson)| Source: Author

The global spatial analysis of log-transformed mean income per person (log_incomeperperson) yielded a very strong, positive regional spatial autocorrelation, with an observed Global Moran's I value of **0.7171**. To test the statistical significance of this finding, a conditional permutation procedure with 999 randomizations was executed to generate a reference distribution under the null hypothesis of complete spatial randomness. The permutation test generated a standard score (**z-score**) of **28.7686** and a highly significant pseudo **p-value of 0.001**.

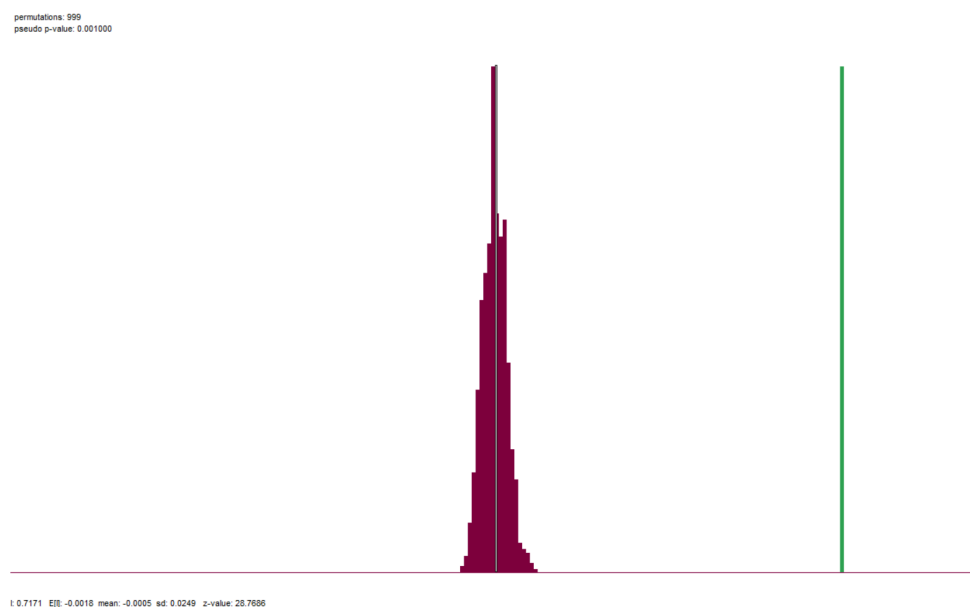


Figure 18: Permutation reference distribution (999 randomizations) for univariate Global Moran's I for income per person per DeSO (log_incomeperperson) | Source: Author.

As shown in the reference distribution histogram, the randomized iterations form a tight bell curve centered near zero. The actual observed Moran's I line (0.7171) sits entirely outside this curve, far to the right of the maximum simulated

value. This result robustly rejects the null hypothesis of spatial randomness ($p = 0.001$), mathematically proving that income distribution across Stockholm is highly segregated and structured into macro-regional clusters rather than an integrated socioeconomic mix.

4.1.3 Univariate LISA for transit density(ln_stops_p1k)

The LISA analysis for public transit density (ln_stops_p1k) identified statistically significant spatial patterns across the study area, breaking down into four distinct typologies:

High-High Clusters (107 units): Highly accessible transit nodes heavily concentrated within Stockholm's urban core and stretching along major structural transit axes to the west.

Low-Low Clusters (50 units): Localized *transit deserts* with low per-capita transit access, predominantly located in the outer northern peripheral districts and southern boundary zones.

Low-High Outliers (19 units): Isolated pockets of under-service consisting of low-transit neighborhoods surrounded by high-density corridors.

High-Low Outliers (7 units): Isolated high-density transit nodes entirely surrounded by low-density areas.

The dominance of 157 synchronized cluster units (High-High and Low-Low) relative to the 26 spatial outliers confirms that local public transport access across Stockholm forms highly cohesive geographic clusters rather than fragmented patterns. This spatial distribution is empirically validated by the underlying significance map, which reaches a pseudo p-value tier of 0.001 across 16 core structural units.

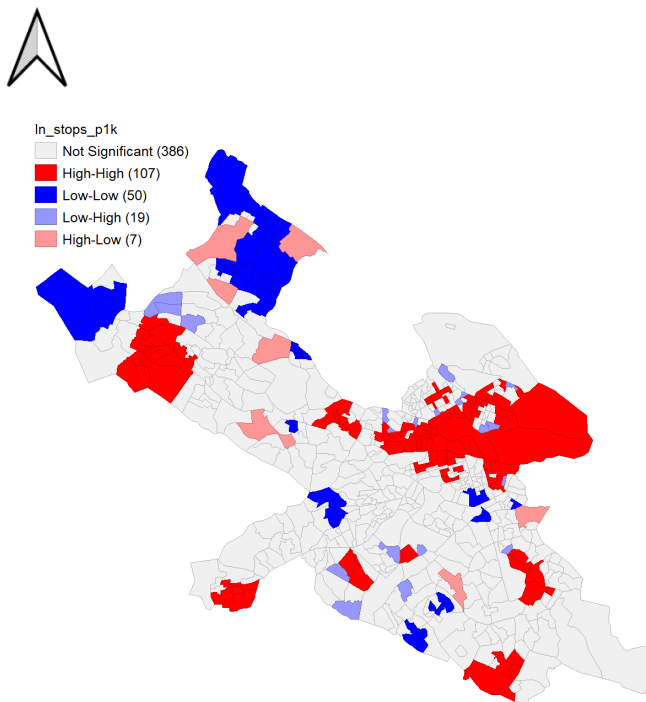


Figure 19: Cluster map for Univariate LISA for(ln_stops_p1k)| Source: Author

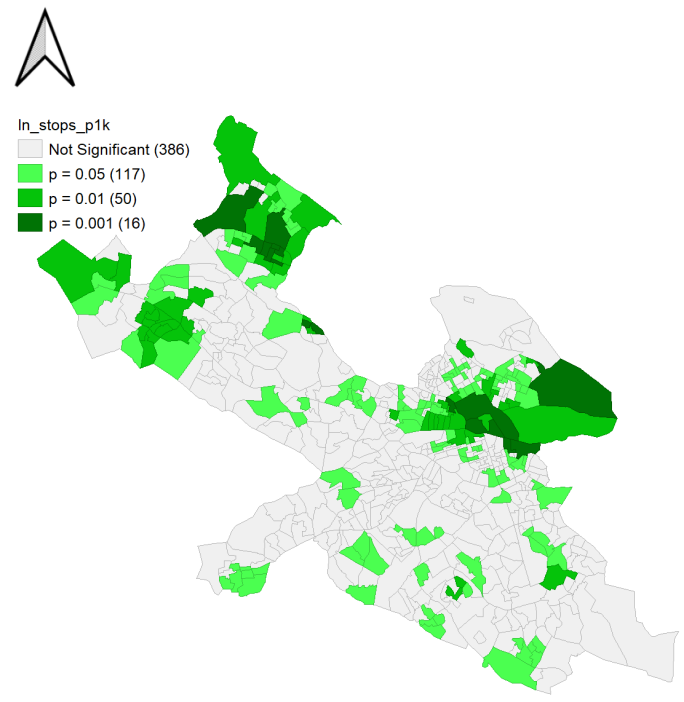


Figure 20: Significance map for Univariate LISA for (ln_stops_p1k)| Source: Author

4.1.4 Univariate LISA for income per person per DeSO (log_incomeperperson)

The LISA analysis for mean per-capita income (log_incomeperperson) identifies strong, highly segregated localized wealth patterns across the study area, breaking down into four distinct typologies:

High-High Clusters (89 units): Concentrated pockets of affluence centered heavily within Stockholm's inner urban core and expanding continuously along the eastern coastal and northern suburban sectors.

Low-Low Clusters (108 units): The largest category of spatial significance, representing clusters of lower-income neighborhoods consolidated within the outer northern peripheral districts and large contiguous sectors of the south.

Low-High Outliers (2 units): Highly rare transitional pockets consisting of lower-income neighborhoods directly isolated within affluent surrounding areas.

High-Low Outliers (1 unit): A singular, highly isolated affluent suburban enclave surrounded by lower-income neighborhoods.

The clear dominance of 197 synchronized clustering units (High-High and Low-Low) versus only 3 spatial outliers underscores a stark regional socio-spatial polarization. This spatial structure is confirmed by the accompanying significance map, which reveals extensive, robust clusters reaching the most conservative pseudo p-value tier ($p = 0.001$) across 65 core structural units.

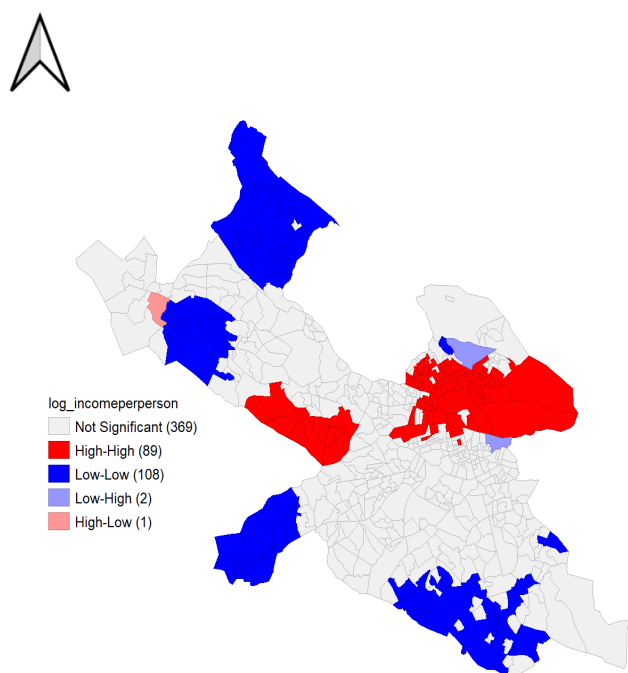


Figure 21: Cluster map for Univariate LISA for log_incomeperperson| Source: Author

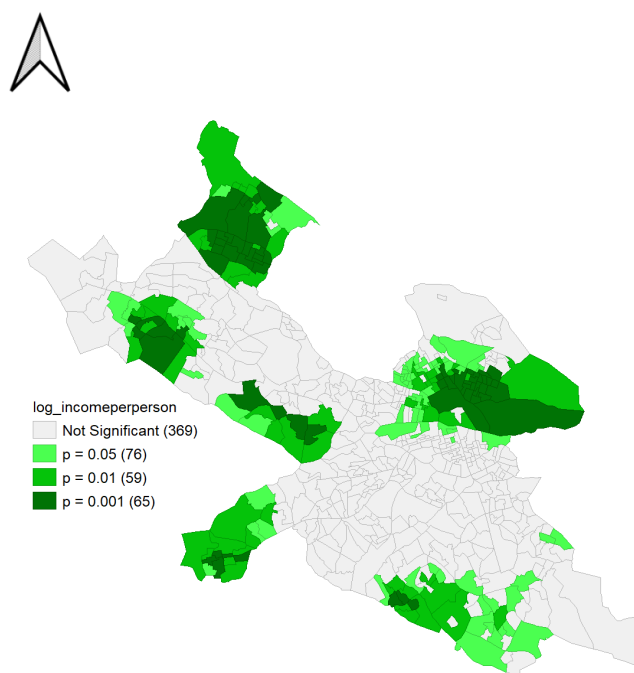


Figure 22: Significance map for Univariate LISA for log_incomeperperson| Source: Author

Bivariate Global Moran's I (log_incomeperperson vs. ln_stops_p1k)

The bivariate global spatial analysis evaluating the overarching relationship between neighborhood wealth and transit density across Stockholm yielded a positive regional spatial autocorrelation, returning an observed Moran's I value of 0.231.

4.1.5 Bivariate Global Moran's I (log_incomeperperson vs. ln_stops_p1k)

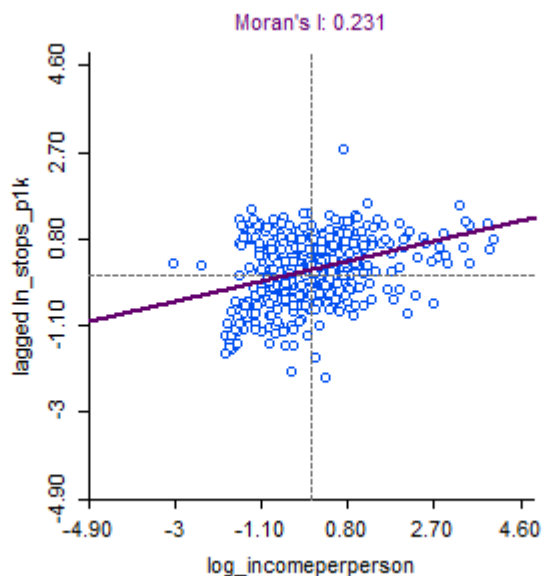
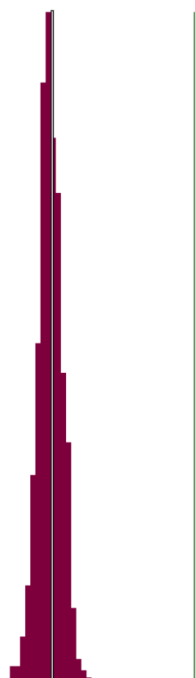


Figure 23: Global Moran's I scatter plot for income per person per DeSO (log_incomeperperson)| Source: Author

The bivariate global spatial analysis evaluating the overarching relationship between neighborhood wealth and transit density across Stockholm yielded a positive regional spatial autocorrelation, returning an observed Moran's I value of **0.231**.

Run

permutations: 999
pseudo p-value: 0.001000



I: 0.2311 E[]: -0.0018 mean: -0.0011 sd: 0.0184 z-value: 12.6025

Figure 24: Permutation reference distribution (999 randomizations) for univariate Global Moran's I for income per person per DeSO (ln_incomeperperson) | Source: Author.

To test the statistical significance of this finding, a conditional permutation procedure with 999 randomizations was executed to generate a reference distribution under the null hypothesis of complete spatial randomness. The permutation test generated a standard score (**z-score**) of **28.7686** and a highly significant pseudo **p-value of 0.001**. As shown in the reference distribution histogram, the randomized iterations form a tight bell curve centered near zero ($E[I] = -0.0018$). The actual observed Moran's I line (0.7171) sits entirely outside this curve, far to the right of the maximum simulated value. This result robustly rejects the null hypothesis of spatial randomness ($p = 0.001$), proving that income distribution across Stockholm is highly segregated and structured into macro-regional clusters.

4.1.6 Bivariate Local Indicators of Spatial Association (BiLISA)

The Bivariate LISA analysis cross-referencing per-capita income against the spatial lag of public transit stop density identified distinct, localized socio-spatial patterns across the metropolitan area, separating into four typologies:

High-High Clusters (79 units): Affluent neighborhoods directly surrounded by high-density public transit networks, heavily concentrated within Stockholm's urban core and inner northeastern suburban sectors.

Low-Low Clusters (48 units): Pockets of compounded disadvantage, representing lower-income neighborhoods surrounded by low-density transit provisions, primarily located in the outer northern peripheral zones.

Low-High Outliers (47 units): Transitional areas where lower-income neighborhoods are positioned within highly accessible, well-connected regional transit corridors.

High-Low Outliers (9 units): Affluent suburban enclaves surrounded by low-density public transit provisions, indicating localized areas likely reliant on private vehicle infrastructure.

The spatial distribution is empirically validated by the accompanying bivariate significance map, which demonstrates high local robustness by reaching a conservative significance tier ($p = 0.001$) across 16 core structural units.

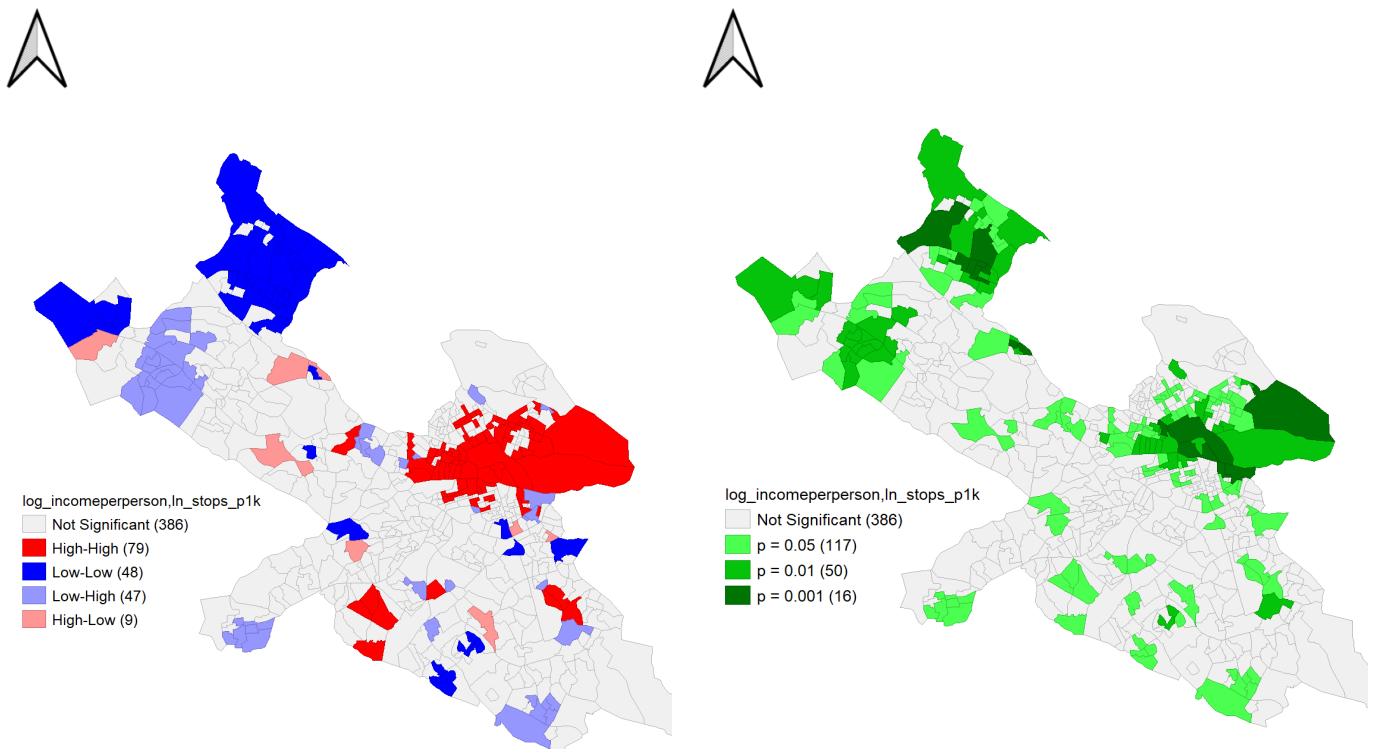


Figure 25: Cluster map for BiLISA| Source: Author

Figure 26: Significance map for BiLISA| Source: Author

4.2 Correlation analysis

Spearman rank correlation analysis was conducted as a preliminary exploratory step to assess the relationships between transit stop density (ln_stops_p1k) and the selected explanatory variables, as well as to identify potential multicollinearity issues prior to regression modelling. As a non-parametric measure, Spearman's correlation was considered appropriate for examining monotonic relationships among socio-spatial variables without assuming normality.

The results indicate statistically significant relationships between transit stop density and several explanatory variables. Income per person exhibited a moderate positive correlation with transit stop density ($\rho = 0.280$, $p < 0.01$), while distance from the CBD showed a moderate negative correlation ($\rho = -0.267$, $p < 0.01$). Car ownership also demonstrated a positive relationship with transit stop density, although the association was comparatively weaker ($\rho = 0.142$, $p < 0.01$). In contrast, the dominant age group variable displayed a weak and statistically insignificant relationship with transit stop density ($\rho = 0.069$, $p = 0.10$).

The Spearman correlation analysis revealed a strong negative association between income per person and distance from the CBD ($\rho = -0.719$, $p < 0.01$), indicating substantial overlap between the two variables. This suggests that both variables capture a similar underlying spatial gradient, whereby higher-income areas tend to be located closer to the city centre. To reduce the risk of multicollinearity and improve model interpretability, distance from the CBD was excluded from the subsequent regression analysis. Income per person was retained because it was more directly aligned with the study's objective of examining the relationship between socioeconomic conditions and transit stop density.

Correlations						
			dom_age_num	log_incomeperperson	carsperson20251_CarsPerson	ln_stops_p1k
Spearman's rho	dom_age_num	Correlation Coefficient	1,000	,084*	,051	-,198**
		Sig. (2-tailed)	.	,044	,221	<,001
		N	569	569	569	569
	log_incomeperperson	Correlation Coefficient	,084*	1,000	,175**	-,719**
		Sig. (2-tailed)	,044	.	<,001	<,001
		N	569	569	569	569
	carsperson20251_CarsPerson	Correlation Coefficient	,051	,175**	1,000	,333**
		Sig. (2-tailed)	,221	<,001	.	<,001
		N	569	569	569	569
	dist_cbd	Correlation Coefficient	-,198**	-,719**	,333**	1,000
		Sig. (2-tailed)	<,001	<,001	<,001	.
		N	569	569	569	569
	ln_stops_p1k	Correlation Coefficient	,069	,280**	,142**	-,267**
		Sig. (2-tailed)	,100	<,001	<,001	<,001
		N	569	569	569	569

*. Correlation is significant at the 0.05 level (2-tailed).
 **. Correlation is significant at the 0.01 level (2-tailed).

Spearman's rho Correlations

Highly Positive: (None)

Positive: (dom_age_num <---> log_incomeperperson), (dom_age_num <---> carsperson20251_CarsPerson), (dom_age_num <---> ln_stops_plk), (log_incomeperperson <---> carsperson20251_CarsPerson), (log_incomeperperson <---> ln_stops_plk), (carsperson20251_CarsPerson <---> dist_cbd), (carsperson20251_CarsPerson <---> ln_stops_plk)

No Linear Correlation: (None)

Negative: (dom_age_num <---> dist_cbd), (log_incomeperperson <---> dist_cbd), (dist_cbd <---> ln_stops_plk)

Highly Negative: (None)

Note: Curated Help is calculated based on actual cell values, not the formatted values.

Figure 27: Spearman correlation analysis| Source: Authors

4.3 Regression analysis

4.3.1 Classical regression analysis in GeoDa

The OLS model explained approximately 8.1% of the variation in transit stop density ($R^2 = 0.081$; Adjusted $R^2 = 0.076$). Although the model was statistically significant overall ($F = 16.65$, $p < 0.001$), its relatively low explanatory power suggests that additional factors influencing transit stop density were not captured by the selected variables. Diagnostic tests indicated no evidence of heteroskedasticity; however, significant spatial autocorrelation remained in the residuals, suggesting that the OLS assumptions were violated and that a spatial regression approach may be more appropriate.

```
>>06/05/26 16:55:47
REGRESSION
-----
SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES ESTIMATION
Data set      : dominant_age_groups
Dependent Variable : ln_stops_plk Number of Observations: 569
Mean dependent var : 1.95544 Number of Variables : 4
S.D. dependent var : 0.397992 Degrees of Freedom : 565

R-squared      : 0.081230 F-statistic      : 16.6509
Adjusted R-squared : 0.076352 Prob(F-statistic) : 2.22687e-10
Sum squared residual: 82.807 Log likelihood : -259.04
Sigma-square    : 0.146561 Akaike info criterion : 526.08
S.E. of regression : 0.382833 Schwarz criterion : 543.456
Sigma-square ML : 0.145531
S.E. of regression ML: 0.381485

-----
Variable      Coefficient      Std. Error      t-Statistic      Probability
-----
CONSTANT      0.469017      0.245274      1.91222      0.05635
dom_age_num    0.00159127    0.000897752    2.21807      0.02695
log_inc..perperson 0.214816    0.0427709    5.02247      0.00000
cars-pe..rs/person 0.373372    0.23746      1.57236      0.11643
-----

REGRESSION DIAGNOSTICS
MULTICOLLINEARITY CONDITION NUMBER 43.309484
TEST ON NORMALITY OF ERRORS
TEST      DF      VALUE      PROB
Jarque-Bera 2      59.6857      0.00000

DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST      DF      VALUE      PROB
Breusch-Pagan test 3      3.1763      0.36523
Koenker-Bassett test 3      1.7715      0.62115

DIAGNOSTICS FOR SPATIAL DEPENDENCE
FOR WEIGHT MATRIX : dominant_age_groups
(row-standardized weights)
TEST      MI/DF      VALUE      PROB
Moran's I (error) 0.3508      14.1905      0.00000
Lagrange Multiplier (lag) 1      194.1480      0.00000
Robust LM (lag) 1      3.1987      0.07370
Lagrange Multiplier (error) 1      192.0819      0.00000
Robust LM (error) 1      1.1325      0.28723
Lagrange Multiplier (SARMA) 2      195.2806      0.00000
=====
END OF REPORT
=====
```

Figure 28: Classic regression model in GeoDa| Source: Authors

Key Coefficient Estimates

Variable	Coefficient (β)	p-value	Significance	Interpretation
Dominant Age Group	0.00199	0.027	Significant	
Income per Person	0.21482	<0.001	Highly Significant	Higher-income areas tend to have higher transit stop density.
Car Ownership	0.37337	0.116	Not Significant	No statistically significant relationship was identified between car ownership and transit stop density.

Spatial Diagnostics

Diagnostic	Value	p-value	Interpretation
Moran's I (Residuals)	0.351	<0.001	Significant spatial autocorrelation remains in the residuals.
LM Lag	194.148	<0.001	Indicates potential spatial dependence.
Robust LM Lag	3.199	0.074	Marginally significant.
LM Error	192.082	<0.001	Indicates potential spatial error dependence.
Robust LM Error	1.133	0.287	Not significant.
LM SARMA	195.281	<0.001	Indicates the presence of spatial dependence.

Because the Lagrange Multiplier (lag) value is slightly higher (194.1480) than the Lagrange Multiplier (error) value (192.0819) we ran the Spatial Lag Model.

4.3.3 Spatial error model regression analysis in GeoDa

Following the OLS analysis, both Spatial Lag and Spatial Error specifications were estimated to account for the significant spatial dependence identified in the residual diagnostics. We estimated both models, which allowed comparison between a process in which transit density is directly influenced by neighbouring areas (SLM) and a process in which spatial dependence arises from omitted spatially structured factors (SEM).

Spatial Lag Model (SLM) Fit

The Spatial Lag Model substantially improved model performance compared with the OLS model by explicitly accounting for spatial dependence between neighbouring DeSOs. The explanatory power increased from $R^2 = 0.081$ in the OLS model to $R^2 = 0.337$ in the SLM, while the Log-likelihood improved from -259.04 to -187.79 and the AIC decreased from 526.08 to 385.58. The spatial autoregressive coefficient ($\rho = 0.593$, $p < 0.001$) was highly significant, indicating that transit stop density in one DeSO is strongly associated with transit stop density in neighbouring areas. Furthermore, the significant Likelihood Ratio Test ($p < 0.001$) confirms that the inclusion of the spatial lag term significantly improves model fit relative to the OLS specification.

```

>>06/05/26 17:10:07
REGRESSION
-----
SUMMARY OF OUTPUT: SPATIAL LAG MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set      : dominant_age_groups
Spatial Weight : dominant_age_groups
Dependent Variable : ln_stops_plk Number of Observations: 569
Mean dependent var : 1.95544 Number of Variables : 5
S.D. dependent var : 0.397992 Degrees of Freedom : 564
Lag coeff. (Rho) : 0.592917

R-squared      : 0.336613 Log likelihood : -187.788
Sq. Correlation : - Akaike info criterion : 385.575
Sigma-square   : 0.105079 Schwarz criterion : 407.295
S.E of regression : 0.324159

-----
Variable      Coefficient      Std.Error      z-value      Probability
-----
W_ln_stops_p  0.592917      0.0438161     13.5319     0.00000
CONSTANT      0.0886507     0.210973      0.420199     0.67434
dom_age_num   0.00122111    0.000760539    1.60559     0.10836
log_inc..perperson 0.0801424    0.0367578     2.18028     0.02924
cars-pe..rs/person 0.553367     0.202332      2.73494     0.00624
-----

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST      DF      VALUE      PROB
Breusch-Pagan test      3      2.4706     0.48062

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL LAG DEPENDENCE FOR WEIGHT MATRIX : dominant_age_groups
TEST      DF      VALUE      PROB
Likelihood Ratio Test    1     142.5047     0.00000
===== END OF REPORT =====

```

Figure 29: Spatial error regression model in Geoda| Source: Authors

Spatial Error Model (SEM) Fit

The Spatial Error Model provided the strongest overall model fit. The explanatory power increased further to $R^2 = 0.355$, while the Log-likelihood improved to -183.97 and the AIC decreased to 375.95. The spatial error coefficient was highly significant ($\lambda = 0.641$, $p < 0.001$), indicating that a substantial proportion of the unexplained variation in transit stop density was spatially structured. The significant Likelihood Ratio Test (LR = 150.13, $p < 0.001$) confirmed that accounting for spatial dependence through the error term significantly improved model performance relative to the OLS model.

```

>>06/05/26 17:10:26
REGRESSION
-----
SUMMARY OF OUTPUT: SPATIAL ERROR MODEL - MAXIMUM LIKELIHOOD ESTIMATION
Data set      : dominant_age_groups
Spatial Weight : dominant_age_groups
Dependent Variable : ln_stops_plk Number of Observations: 569
Mean dependent var : 1.955441 Number of Variables : 4
S.D. dependent var : 0.397992 Degrees of Freedom : 565
Lag coeff. (Lambda) : 0.640775

R-squared      : 0.355479 R-squared (BUSE) : -
Sq. Correlation : - Log likelihood : -183.972746
Sigma-square   : 0.10209 Akaike info criterion : 375.945
S.E of regression : 0.319516 Schwarz criterion : 393.321

-----
Variable      Coefficient      Std.Error      z-value      Probability
-----
CONSTANT      1.36818      0.365359      3.74474     0.00018
dom_age_num   0.000660033  0.000758153    0.87058     0.38398
log_inc..perperson 0.0199566    0.0649723     0.307156     0.75873
cars-pe..rs/person 1.39503      0.318418      4.38113     0.00001
LAMBDA        0.640775     0.0415481     15.4225     0.00000
-----

REGRESSION DIAGNOSTICS
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST      DF      VALUE      PROB
Breusch-Pagan test      3      1.5363     0.67392

DIAGNOSTICS FOR SPATIAL DEPENDENCE
SPATIAL ERROR DEPENDENCE FOR WEIGHT MATRIX : dominant_age_groups
TEST      DF      VALUE      PROB
Likelihood Ratio Test    1     150.1346     0.00000
===== END OF REPORT =====

```

Figure 30: Spatial error regression model in Geoda| Source: Authors

Key coefficient estimates

Model	R ²	Log-Likelihood	AIC	Spatial Parameter
OLS	0.081	-259.04	526.08	–
SLM	0.337	-187.79	385.58	$\rho = 0.593$
SEM	0.355	-183.97	375.95	$\lambda = 0.641$

5. Discussion

The LISA and Moran's I analysis showed that both transit density and income are spatially clustered across Stockholm rather than randomly distributed. High-high transit density clusters were mainly concentrated around the inner-city areas such as Norrmalm, Vasastan, Östermalm and parts of Södermalm, while low-low accessibility clusters appeared more frequently within peripheral northern and southern districts. Similarly, the income LISA showed affluent clusters concentrated in areas such as Östermalm, Bromma and Djurgården, while lower-income clusters were visible in districts such as Farsta, Skärholmen, Rinkeby, Tensta and Husby.

However, the BiLISA results showed that the relationship between income and transit density was not entirely straightforward. While some lower-income peripheral districts showed weaker transit density patterns, several lower-income areas also remained relatively well connected due to their location along Stockholm's metro corridors and higher dependence on public transport. At the same time, affluent inner-city districts such as Norrmalm and Östermalm combined both high income and high transit density, while areas such as Bromma showed relatively higher income despite comparatively lower transit density. Therefore, our analysis did not fully confirm the initial assumption that poorer areas consistently experience lower transit density.

We therefore complemented ESDA with statistical methods to quantify these relationships. The combination of Spearman correlation, OLS regression, and spatial regression models allowed us to progressively refine our understanding of the relationships within the dataset. We first used Spearman correlation as an exploratory tool to identify pairwise relationships and potential multicollinearity issues. The strong correlation between income and distance from the CBD suggested substantial overlap between the two variables. Because income was more closely aligned with the study's focus on socioeconomic influences on transit accessibility, we retained income and excluded CBD distance from the subsequent regression models.

We then applied OLS regression to examine the independent effects of the explanatory variables while controlling for the influence of the others. This represented an important methodological step beyond correlation analysis, as variables that appeared important in bivariate relationships did not necessarily remain significant in a multivariate context. However, the presence of significant spatial autocorrelation in the residuals indicated that the OLS assumptions were violated and that spatial processes were not adequately captured.

To address this limitation, we estimated both Spatial Lag and Spatial Error models. Both models substantially improved fit compared with OLS, demonstrating the importance of accounting for spatial dependence when analysing urban accessibility patterns. While the Spatial Lag Model captured potential interactions between neighbouring areas, the Spatial Error Model provided the strongest overall fit, suggesting that spatial clustering in transit stop density is better explained by unobserved spatially structured factors than by direct spillover effects. Overall, the progression from exploratory correlation analysis to spatial econometric modelling highlights the importance of selecting methods that account for both multivariate relationships and spatial dependence in geographic datasets.

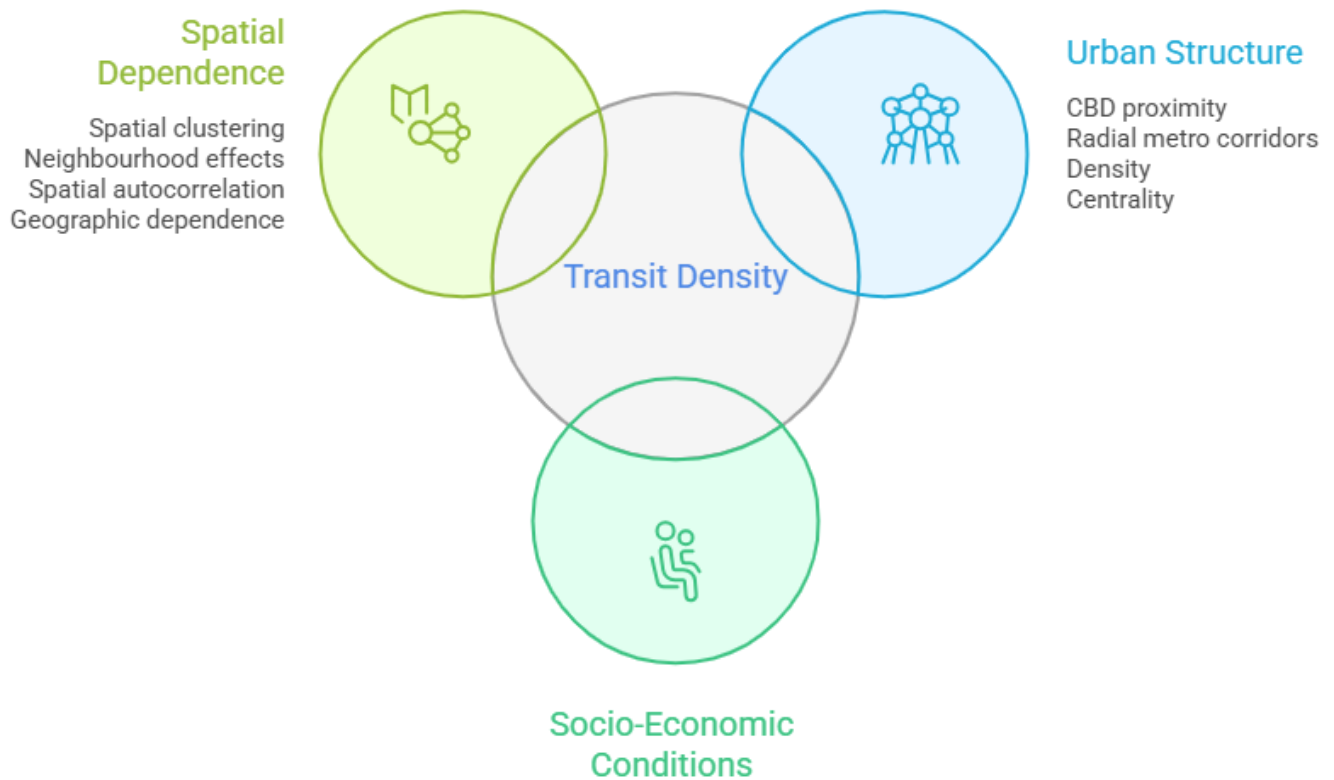


Figure 30: Factors affecting transit accessibility(Density in our case)| Source: Authors

6. Future research questions

Based on the analysis, interpretations, and methodological constraints identified in this study, several opportunities exist to refine both the spatial modeling and the visualization of public transit accessibility. Suggestions for future work on this dataset include:

- Expanding the spatial boundary to the regional scale to eliminate administrative edge-effects and capture continuous cross-border commuting flows.
- Integrating explicit land-use and environmental layers, such as coastal geometry, topography, and protected green infrastructure, to directly model the unobserved physical drivers currently absorbed by the spatial error (lambda) term.
- Adding housing market indicators, such as average rental prices or property values per DeSO, to capture how gentrification, transit-oriented development, and the housing market intersect with infrastructure access.

7. Conclusion

This study evaluated the effectiveness of a sequence of spatial and statistical methods in analysing the spatial distribution of transit density across Stockholm's DeSO areas. We did not want to just focus on explaining transit patterns, we assessed how different methodological approaches shape the interpretation of spatial relationships.

The results show that exploratory spatial analysis (ESDA), correlation analysis, and regression models each capture different dimensions of the same process. ESDA was effective in revealing the presence of spatial clustering in both transit density and income, but it remained descriptive and could not explain underlying relationships. Correlation analysis helped us identify broad associations and potential multicollinearity, but it was limited to pairwise relationships and did not account for spatial structure.

OLS regression provided a more rigorous multivariate framework, but its results were affected by significant spatial autocorrelation in the residuals, demonstrating that conventional regression assumptions are not suitable for spatial urban data. We addressed this limitation through spatial econometric models, which significantly improved model fit and highlighted the importance of accounting for spatial dependence.

Among the tested models, the Spatial Error Model provided the most robust representation of the data structure, suggesting that unobserved spatial processes play a stronger role than direct spatial interactions between neighbouring areas. Importantly, the comparison across methods shows that conclusions about the role of socio-economic variables are sensitive to methodological choice, and may change substantially once spatial dependence is explicitly modelled.

To conclude, our study demonstrates that no single method is sufficient to evaluate spatial distribution in urban systems. Instead, robust analysis requires a layered approach in which exploratory spatial analysis, statistical modelling, and spatial econometrics are used together. The key contribution of this work is therefore methodological: we show how different analytical frameworks can lead to different interpretations of the same urban dataset, and why spatial dependence must be explicitly accounted for in studies of urban accessibility.

Method	Strengths	Limitations
ESDA	Shows spatial clustering and local variation and useful for intuitive visual interpretation	Descriptive only and cannot test causality or control for variables
Spearman Correlation	Identifies initial relationships and potential multicollinearity	Limited to bivariate relationships; ignores spatial dependence
OLS Regression	Enables multivariate analysis and controls for multiple variables	Assumes spatial independence; sensitive to spatial autocorrelation
Spatial Lag Model	Considers potential spillover effects between neighbouring areas	Less suitable when dependence is driven by omitted variables
Spatial Error Model	Best statistical fit for our study and accounts for unobserved spatial structure	Less interpretable in terms of direct spatial interaction

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Appendix

Attribute Name	Description / Operationalization
ln_stops_plk	Dependent Variable (Y): Natural log of public transit stops per 1,000 residents within the DeSO unit (ln_stops_p1k).
log_incomeperperson	Independent Variable (X): Natural log of income per person (log_incomeperperson).
dist_cbd	Control Variable (X): Linear Euclidean distance from the DeSO geometric centroid to Stockholm Central Station (km).
cars-pe..rs/person	Control Variable (X): Registered private vehicles per person